

Question ID 7355b9d9

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 7355b9d9

If  $k - x$  is a factor of the expression  $-x^2 + \frac{1}{29}nk^2$ , where  $n$  and  $k$  are constants and  $k > 0$ , what is the value of  $n$ ?

- A.  $-29$
- B.  $-\frac{1}{29}$
- C.  $\frac{1}{29}$
- D.  $29$

ID: 7355b9d9 Answer

Correct Answer:

D

Rationale

Choice D is correct. If  $k - x$  is a factor of the expression  $-x^2 + \left(\frac{1}{29}\right)nk^2$ , then the expression can be written as  $(k - x)(ax + b)$ , where  $a$  and  $b$  are constants. This expression can be rewritten as  $akx + bk - ax^2 - bx$ , or  $-ax^2 + (ak - b)x + bk$ . Since this expression is equivalent to  $-x^2 + \left(\frac{1}{29}\right)nk^2$ , it follows that  $-a = -1$ ,  $ak - b = 0$ , and  $bk = \left(\frac{1}{29}\right)nk^2$ . Dividing each side of the equation  $-a = -1$  by  $-1$  yields  $a = 1$ . Substituting  $1$  for  $a$  in the equation  $ak - b = 0$  yields  $k - b = 0$ . Adding  $b$  to each side of this equation yields  $k = b$ . Substituting  $k$  for  $b$  in the equation  $bk = \left(\frac{1}{29}\right)nk^2$  yields  $k^2 = \left(\frac{1}{29}\right)nk^2$ . Since  $k$  is positive, dividing each side of this equation by  $k^2$  yields  $1 = \left(\frac{1}{29}\right)n$ . Multiplying each side of this equation by  $29$  yields  $29 = n$ .

Alternate approach: The expression  $x^2 - y^2$  can be written as  $(x - y)(x + y)$ , which is a difference of two squares. It follows that  $\left(\frac{1}{29}\right)nk^2 - x^2$  is equivalent to  $\left(\left(\sqrt{\frac{1}{29}n}\right)k - x\right)\left(\left(\sqrt{\frac{1}{29}n}\right)k + x\right)$ . It's given that  $k - x$  is a factor of  $-x^2 + \left(\frac{1}{29}\right)nk^2$ , so the factor  $\left(\sqrt{\frac{1}{29}n}\right)k - x$  is equal to  $k - x$ . Adding  $x$  to both sides of the equation  $\left(\sqrt{\frac{1}{29}n}\right)k - x = k - x$  yields  $\left(\sqrt{\frac{1}{29}n}\right)k = k$ . Since  $k$  is positive, dividing both sides of this equation by  $k$  yields  $\sqrt{\frac{1}{29}n} = 1$ . Squaring both sides of this equation yields  $\frac{1}{29}n = 1$ . Multiplying both sides of this equation by  $29$  yields  $n = 29$ .

Choice A is incorrect. This value of  $n$  gives the expression  $-x^2 + \left(\frac{1}{29}\right)(-29)k^2$ , or  $-x^2 - k^2$ . This expression doesn't have  $k - x$  as a factor.

Choice B is incorrect. This value of  $n$  gives the expression  $-x^2 + \left(\frac{1}{29}\right)\left(-\frac{1}{29}\right)k^2$ , or  $-x^2 + \left(-\frac{1}{841}\right)k^2$ . This expression doesn't have  $k - x$  as a factor.

Choice C is incorrect. This value of  $n$  gives the expression  $-x^2 + \left(\frac{1}{29}\right)\left(\frac{1}{29}\right)k^2$ , or  $-x^2 + \left(\frac{1}{841}\right)k^2$ . This expression doesn't have  $k - x$  as a factor.

Question Difficulty:

Hard

Question ID e117d3b8

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: e117d3b8

If  $a$  and  $c$  are positive numbers, which of the following is equivalent to  $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$ ?

- A.  $a+c$
- B.  $a^2+c^2$
- C.  $a^2+2ac+c^2$
- D.  $a^2c^2$

ID: e117d3b8 Answer

Correct Answer:  
C

Rationale

Choice C is correct. Using the property that  $\sqrt{x}\sqrt{y} = \sqrt{xy}$  for positive numbers  $x$  and  $y$ , with  $x = (a+c)^3$  and  $y = a+c$ , it follows that  $\sqrt{(a+c)^3} \cdot \sqrt{a+c} = \sqrt{(a+c)^4}$ . By rewriting  $(a+c)^4$  as  $((a+c)^2)^2$ , it is possible to simplify the square root expression as follows:  $\sqrt{((a+c)^2)^2} = (a+c)^2 = a^2+2ac+c^2$ .

Choice A is incorrect and may be the result of  $\sqrt{(a+c)^3} \div \sqrt{(a+c)}$ . Choice B is incorrect and may be the result of incorrectly rewriting  $(a+c)^2$  as  $a^2+c^2$ . Choice D is incorrect and may be the result of incorrectly applying properties of exponents.

Question Difficulty:  
Hard

# Question ID f89e1d6f

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: f89e1d6f

If  $a = c + d$ , which of the following is equivalent to the expression  $x^2 - c^2 - 2cd - d^2$ ?

- A.  $(x + a)^2$
- B.  $(x - a)^2$
- C.  $(x + a)(x - a)$
- D.  $x^2 - ax - a^2$

ID: f89e1d6f Answer

Correct Answer:  
C

Rationale

Choice C is correct. Factoring  $-1$  from the second, third, and fourth terms gives  $x^2 - c^2 - 2cd - d^2 = x^2 - (c^2 + 2cd + d^2)$ . The expression  $c^2 + 2cd + d^2$  is the expanded form of a perfect square:  $c^2 + 2cd + d^2 = (c + d)^2$ . Therefore,  $x^2 - (c^2 + 2cd + d^2) = x^2 - (c + d)^2$ . Since  $a = c + d$ ,  $x^2 - (c + d)^2 = x^2 - a^2$ . Finally, because  $x^2 - a^2$  is the difference of squares, it can be expanded as  $x^2 - a^2 = (x + a)(x - a)$ .

Choices A and B are incorrect and may be the result of making an error in factoring the difference of squares  $x^2 - a^2$ . Choice D is incorrect and may be the result of incorrectly combining terms.

Question Difficulty:  
Hard

# Question ID b74f2feb

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: b74f2feb

The expression  $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$  is equivalent to  $ax^b$ , where  $a$  and  $b$  are positive constants and  $x > 1$ . What is the value of  $a + b$ ?

ID: b74f2feb Answer

**Correct Answer:**  
361/8, 45.12, 45.13

**Rationale**

The correct answer is  $\frac{361}{8}$ . The rational exponent property is  $\sqrt[n]{y^m} = y^{\frac{m}{n}}$ , where  $y > 0$ ,  $m$  and  $n$  are integers, and  $n > 0$ . This property can be applied to rewrite the given expression  $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$  as  $6\left(3^{\frac{5}{5}}\right)\left(x^{\frac{45}{5}}\right)\left(2^{\frac{8}{8}}\right)\left(x^{\frac{1}{8}}\right)$ , or  $6(3)(x^9)(2)\left(x^{\frac{1}{8}}\right)$ . This expression can be rewritten by multiplying the constants, which gives  $36(x^9)\left(x^{\frac{1}{8}}\right)$ . The multiplication exponent property is  $y^n \cdot y^m = y^{n+m}$ , where  $y > 0$ . This property can be applied to rewrite the expression  $36(x^9)\left(x^{\frac{1}{8}}\right)$  as  $36x^{9+\frac{1}{8}}$ , or  $36x^{\frac{73}{8}}$ . Therefore,  $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x} = 36x^{\frac{73}{8}}$ . It's given that  $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$  is equivalent to  $ax^b$ ; therefore,  $a = 36$  and  $b = \frac{73}{8}$ . It follows that  $a + b = 36 + \frac{73}{8}$ . Finding a common denominator on the right-hand side of this equation gives  $a + b = \frac{288}{8} + \frac{73}{8}$ , or  $a + b = \frac{361}{8}$ . Note that 361/8, 45.12, and 45.13 are examples of ways to enter a correct answer.

**Question Difficulty:**  
Hard

Question ID 911c415b

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 911c415b

$(7532 + 100y^2) + 10(10y^2 - 110)$

The expression above can be written in the form  $ay^2 + b$ , where  $a$  and  $b$  are constants. What is the value of  $a + b$ ?

ID: 911c415b Answer

Rationale

The correct answer is 6632. Applying the distributive property to the expression yields  $(7532 + 100y^2) + (100y^2 - 1100)$ . Then adding together  $7532 + 100y^2$  and  $100y^2 - 1100$  and collecting like terms results in  $200y^2 + 6432$ . This is written in the form  $ay^2 + b$ , where  $a = 200$  and  $b = 6432$ . Therefore  $a + b = 200 + 6432 = 6632$ .

Question Difficulty:  
Hard

# Question ID 89fc23af

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 89fc23af

Which of the following expressions is

equivalent to  $\frac{x^2-2x-5}{x-3}$  ?

- A.  $x - 5 - \frac{20}{x-3}$
- B.  $x - 5 - \frac{10}{x-3}$
- C.  $x + 1 - \frac{8}{x-3}$
- D.  $x + 1 - \frac{2}{x-3}$

ID: 89fc23af Answer

Correct Answer:

D

Rationale

Choice D is correct. The numerator of the given expression can be rewritten in terms of the denominator,  $x - 3$ , as follows:

$x^2 - 2x - 5 = x^2 - 3x + x - 3 - 2$ , which is equivalent to  $x(x - 3) + (x - 3) - 2$ . So the given expression is equivalent to  $\frac{x(x - 3) + (x - 3) - 2}{x - 3} = \frac{x(x - 3)}{x - 3} + \frac{x - 3}{x - 3} - \frac{2}{x - 3}$ . Since the given expression is defined for  $x \neq 3$ , the expression can be rewritten as  $x + 1 - \frac{2}{x - 3}$ .

Long division can also be used as an alternate approach. Choices A, B, and C are incorrect and may result from errors made when dividing the two polynomials or making use of structure.

Question Difficulty:

Hard

Question ID 12e7faf8

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 12e7faf8

The equation  $\frac{x^2+6x-7}{x+7} = ax+d$  is true for all  $x \neq -7$ , where  $a$  and  $d$  are integers. What is the value of  $a+d$ ?

- A. -6
- B. -1
- C. 0
- D. 1

ID: 12e7faf8 Answer

Correct Answer:  
C

Rationale

Choice C is correct. Since the expression  $x^2+6x-7$  can be factored as  $(x+7)(x-1)$ , the given equation can be rewritten as  $\frac{(x+7)(x-1)}{x+7} = ax+d$ . Since  $x \neq -7$ ,  $x+7$  is also not equal to 0, so both the numerator and denominator of  $\frac{(x+7)(x-1)}{x+7}$  can be divided by  $x+7$ . This gives  $x-1 = ax+d$ . Equating the coefficient of  $x$  on each side of the equation gives  $a = 1$ . Equating the constant terms gives  $d = -1$ . The sum is  $1+(-1) = 0$ .

Choice A is incorrect and may result from incorrectly simplifying the equation. Choices B and D are incorrect. They are the values of  $d$  and  $a$ , respectively, not  $a+d$ .

Question Difficulty:  
Hard

# Question ID 42f8e4b4

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 42f8e4b4

One of the factors of  $2x^3 + 42x^2 + 208x$  is  $x + b$ , where  $b$  is a positive constant. What is the smallest possible value of  $b$ ?

ID: 42f8e4b4 Answer

Correct Answer:

8

### Rationale

The correct answer is 8. Since each term of the given expression,  $2x^3 + 42x^2 + 208x$ , has a factor of  $2x$ , the expression can be rewritten as  $2x(x^2) + 2x(21x) + 2x(104)$ , or  $2x(x^2 + 21x + 104)$ . Since the values 8 and 13 have a sum of 21 and a product of 104, the expression  $x^2 + 21x + 104$  can be factored as  $(x + 8)(x + 13)$ . Therefore, the given expression can be factored as  $2x(x + 8)(x + 13)$ . It follows that the factors of the given expression are 2,  $x$ ,  $x + 8$ , and  $x + 13$ . Of these factors, only  $x + 8$  and  $x + 13$  are of the form  $x + b$ , where  $b$  is a positive constant. Therefore, the possible values of  $b$  are 8 and 13. Thus, the smallest possible value of  $b$  is 8.

Question Difficulty:

Hard



# Question ID 20291f47

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 20291f47

Which expression is equivalent to  $\frac{y+12}{x-8} + \frac{y(x-8)}{x^2y-8xy}$ ?

- A.  $\frac{xy+y+4}{x^3y-16x^2y+64xy}$
- B.  $\frac{xy+9y+12}{x^2y-8xy+x-8}$
- C.  $\frac{xy^2+13xy-8y}{x^2y-8xy}$
- D.  $\frac{xy^2+13xy-8y}{x^3y-16x^2y+64xy}$

ID: 20291f47 Answer

Correct Answer:

C

### Rationale

Choice C is correct. Factoring the denominator in the second term of the given expression gives  $\frac{y+12}{x-8} + \frac{y(x-8)}{xy(x-8)}$ . This expression can be rewritten with common denominators by multiplying the first term by  $\frac{xy}{xy}$ , giving  $\frac{xy(y+12)}{xy(x-8)} + \frac{y(x-8)}{xy(x-8)}$ . Adding these two terms yields  $\frac{xy(y+12)+y(x-8)}{xy(x-8)}$ . Using the distributive property to rewrite this expression gives  $\frac{xy^2+12xy+xy-8y}{x^2y-8xy}$ . Combining the like terms in the numerator of this expression gives  $\frac{xy^2+13xy-8y}{x^2y-8xy}$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID ad038c19

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: ad038c19

Which of the following is

equivalent to  $\left(a + \frac{b}{2}\right)^2$  ?

A.  $a^2 + \frac{b^2}{2}$

B.  $a^2 + \frac{b^2}{4}$

C.  $a^2 + \frac{ab}{2} + \frac{b^2}{2}$

D.  $a^2 + ab + \frac{b^2}{4}$

ID: ad038c19 Answer

Correct Answer:

D

Rationale

Choice D is correct. The expression  $\left(a + \frac{b}{2}\right)^2$  can be rewritten as  $\left(a + \frac{b}{2}\right)\left(a + \frac{b}{2}\right)$ . Using the distributive property, the expression yields  $\left(a + \frac{b}{2}\right)\left(a + \frac{b}{2}\right) = a^2 + \frac{ab}{2} + \frac{ab}{2} + \frac{b^2}{4}$ . Combining like terms gives  $a^2 + ab + \frac{b^2}{4}$ .

Choices A, B, and C are incorrect and may result from errors using the distributive property on the given expression or combining like terms.

Question Difficulty:

Hard

# Question ID c6e85cd7

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: c6e85cd7

If  $4^{8c} = \sqrt[3]{4^7}$ , what is the value of  $c$ ?

ID: c6e85cd7 Answer

**Correct Answer:**  
.2916, .2917, 7/24

**Rationale**

The correct answer is  $\frac{7}{24}$ . An expression of the form  $\sqrt[n]{a^m}$ , where  $m$  and  $n$  are integers greater than 1 and  $a \geq 0$ , is equivalent to  $a^{\frac{m}{n}}$ . Therefore, the expression on the right-hand side of the given equation,  $\sqrt[3]{4^7}$ , is equivalent to  $4^{\frac{7}{3}}$ . Thus,  $4^{8c} = 4^{\frac{7}{3}}$ . It follows that  $8c = \frac{7}{3}$ . Dividing both sides of this equation by 8 yields  $c = \frac{7}{24}$ . Note that 7/24, .2916, .2917, 0.219, and 0.292 are examples of ways to enter a correct answer.

**Question Difficulty:**  
Hard

Question ID a0b4103e

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: a0b4103e

The expression  $\frac{1}{3}x^2 - 2$  can be rewritten as  $\frac{1}{3}(x - k)(x + k)$ , where  $k$  is a positive constant. What is the value of  $k$  ?

- A. 2
- B. 6
- C.  $\sqrt{2}$
- D.  $\sqrt{6}$

ID: a0b4103e Answer

Correct Answer:  
D

Rationale

Choice D is correct. Factoring out the coefficient  $\frac{1}{3}$ , the given expression can be rewritten as  $\frac{1}{3}(x^2 - 6)$ . The expression  $x^2 - 6$  can be approached as a difference of squares and rewritten as  $(x - \sqrt{6})(x + \sqrt{6})$ . Therefore,  $k$  must be  $\sqrt{6}$ .

Choice A is incorrect. If  $k$  were 2, then the expression given would be rewritten as  $\frac{1}{3}(x - 2)(x + 2)$ , which is equivalent to  $\frac{1}{3}x^2 - \frac{4}{3}$ , not  $\frac{1}{3}x^2 - 2$ .

Choice B is incorrect. This may result from incorrectly factoring the expression and finding  $(x - 6)(x + 6)$  as the factored form of the expression. Choice C is incorrect. This may result from incorrectly distributing the  $\frac{1}{3}$  and rewriting the expression as  $\frac{1}{3}(x^2 - 2)$ .

Question Difficulty:  
Hard

Question ID 22fd3e1f

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 22fd3e1f

$$f(x) = x^3 - 9x$$
$$g(x) = x^2 - 2x - 3$$

Which of the following expressions is

equivalent to  $\frac{f(x)}{g(x)}$ , for  $x > 3$ ?

- A.  $\frac{1}{x+1}$
- B.  $\frac{x+3}{x+1}$
- C.  $\frac{x(x-3)}{x+1}$
- D.  $\frac{x(x+3)}{x+1}$

ID: 22fd3e1f Answer

Correct Answer:

D

Rationale

Choice D is correct. Since  $x^3 - 9x = x(x+3)(x-3)$  and  $x^2 - 2x - 3 = (x+1)(x-3)$ , the fraction  $\frac{f(x)}{g(x)}$  can be written as  $\frac{x(x+3)(x-3)}{(x+1)(x-3)}$ . It is given that  $x > 3$ , so the common factor  $x - 3$  is not equal to 0. Therefore, the fraction can be further simplified to  $\frac{x(x+3)}{x+1}$ .

Choice A is incorrect. The expression  $\frac{1}{x+1}$  is not equivalent to  $\frac{f(x)}{g(x)}$  because at  $x = 0$ ,  $\frac{1}{x+1}$  has a value of 1 and  $\frac{f(x)}{g(x)}$  has a value of 0.

Choice B is incorrect and results from omitting the factor x in the factorization of  $f(x)$ . Choice C is incorrect and may result from incorrectly factoring  $g(x)$  as  $(x+1)(x+3)$  instead of  $(x+1)(x-3)$ .

**Question Difficulty:**  
Hard

# Question ID ffdbcad4

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: ffdbcad4

The expression  $4x^2 + bx - 45$ , where  $b$  is a constant, can be rewritten as  $(hx + k)(x + j)$ , where  $h$ ,  $k$ , and  $j$  are integer constants. Which of the following must be an integer?

- A.  $\frac{b}{h}$
- B.  $\frac{b}{k}$
- C.  $\frac{45}{h}$
- D.  $\frac{45}{k}$

ID: ffdbcad4 Answer

Correct Answer:  
D

### Rationale

Choice D is correct. It's given that  $4x^2 + bx - 45$  can be rewritten as  $(hx + k)(x + j)$ . The expression  $(hx + k)(x + j)$  can be rewritten as  $hx^2 + jhx + kx + kj$ , or  $hx^2 + (jh + k)x + kj$ . Therefore,  $hx^2 + (jh + k)x + kj$  is equivalent to  $4x^2 + bx - 45$ . It follows that  $kj = -45$ . Dividing each side of this equation by  $k$  yields  $j = \frac{-45}{k}$ . Since  $j$  is an integer,  $-\frac{45}{k}$  must be an integer. Therefore,  $\frac{45}{k}$  must also be an integer.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:  
Hard

Question ID 2c88af4d

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 2c88af4d

$$\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$$

The expression  $x^{\frac{1}{3}}y^{-1}$ , where  $x > 1$  and  $y > 1$ , is equivalent to which of the following?

- $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

A.
- $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

B.
- $\frac{y\sqrt{y}}{x\sqrt{x}}$

C.
- $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

D.

ID: 2c88af4d Answer

Correct Answer:  
D

Rationale

Choice D is correct. For  $x > 1$  and  $y > 1$ ,  $x^{\frac{1}{3}}$  and  $y^{\frac{1}{2}}$  are equivalent to  $\sqrt[3]{x}$  and  $\sqrt{y}$ , respectively. Also,  $x^{-2}$  and  $y^{-1}$  are equivalent to  $\frac{1}{x^2}$  and  $\frac{1}{y}$ , respectively. Therefore, the given expression can be rewritten as  $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$ .

Choices A, B, and C are incorrect because these choices are not equivalent to the given expression for  $x > 1$  and  $y > 1$ .

For example, for  $x = 2$  and  $y = 2$ , the value of the given expression is  $2^{-\frac{5}{6}}$ ; the values of the choices, however, are  $2^{-\frac{1}{3}}$ ,  $2^{\frac{5}{6}}$ , and 1, respectively.

Question Difficulty:  
Hard



# Question ID 967ef685

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 967ef685

Which expression is equivalent to  $\frac{42a}{k} + 42ak$ , where  $k > 0$ ?

- A.  $\frac{84a}{k}$
- B.  $\frac{84ak^2}{k}$
- C.  $\frac{42a(k+1)}{k}$
- D.  $\frac{42a(k^2+1)}{k}$

ID: 967ef685 Answer

Correct Answer:

D

### Rationale

Choice D is correct. Two fractions can be added together when they have a common denominator. Since  $k > 0$ , multiplying the second term in the given expression by  $\frac{k}{k}$  yields  $\frac{(42ak)k}{k}$ , which is equivalent to  $\frac{42ak^2}{k}$ . Therefore, the expression  $\frac{42a}{k} + 42ak$  can be written as  $\frac{42a}{k} + \frac{42ak^2}{k}$  which is equivalent to  $\frac{42a+42ak^2}{k}$ . Since each term in the numerator of this expression has a factor of  $42a$ , the expression  $\frac{42a+42ak^2}{k}$  can be rewritten as  $\frac{42a(1)+42a(k^2)}{k}$ , or  $\frac{42a(1+k^2)}{k}$ , which is equivalent to  $\frac{42a(k^2+1)}{k}$ .

Choice A is incorrect. This expression is equivalent to  $\frac{42a}{k} + \frac{42a}{k}$ .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This expression is equivalent to  $\frac{42a}{k} + 42a$ .

Question Difficulty:

Hard

# Question ID e51bf5b1

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: e51bf5b1

Which of the following expressions has a factor of  $x + 2b$ , where  $b$  is a positive integer constant?

- A.  $3x^2 + 7x + 14b$
- B.  $3x^2 + 28x + 14b$
- C.  $3x^2 + 42x + 14b$
- D.  $3x^2 + 49x + 14b$

ID: e51bf5b1 Answer

Correct Answer:

D

### Rationale

Choice D is correct. Since each choice has a term of  $3x^2$ , which can be written as  $(3x)(x)$ , and each choice has a term of  $14b$ , which can be written as  $(7)(2b)$ , the expression that has a factor of  $x + 2b$ , where  $b$  is a positive integer constant, can be represented as  $(3x + 7)(x + 2b)$ . Using the distributive property of multiplication, this expression is equivalent to  $3x(x + 2b) + 7(x + 2b)$ , or  $3x^2 + 6xb + 7x + 14b$ . Combining the  $x$ -terms in this expression yields  $3x^2 + (7 + 6b)x + 14b$ . It follows that the coefficient of the  $x$ -term is equal to  $7 + 6b$ . Thus, from the given choices,  $7 + 6b$  must be equal to  $7, 28, 42$ , or  $49$ . Therefore,  $6b$  must be equal to  $0, 21, 35$ , or  $42$ , respectively, and  $b$  must be equal to  $\frac{0}{6}, \frac{21}{6}, \frac{35}{6}$ , or  $\frac{42}{6}$ , respectively. Of these four values of  $b$ , only  $\frac{42}{6}$ , or  $7$ , is a positive integer. It follows that  $7 + 6b$  must be equal to  $49$  because this is the only choice for which the value of  $b$  is a positive integer constant. Therefore, the expression that has a factor of  $x + 2b$  is  $3x^2 + 49x + 14b$ .

Choice A is incorrect. If this expression has a factor of  $x + 2b$ , then the value of  $b$  is  $0$ , which isn't positive.

Choice B is incorrect. If this expression has a factor of  $x + 2b$ , then the value of  $b$  is  $\frac{21}{6}$ , which isn't an integer.

Choice C is incorrect. If this expression has a factor of  $x + 2b$ , then the value of  $b$  is  $\frac{35}{6}$ , which isn't an integer.

### Question Difficulty:

Hard

# Question ID c81b6c57

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: c81b6c57

In the expression  $3(2x^2 + px + 8) - 16x(p + 4)$ ,  $p$  is a constant. This expression is equivalent to the expression  $6x^2 - 155x + 24$ . What is the value of  $p$  ?

- A.  $-3$
- B.  $7$
- C.  $13$
- D.  $155$

ID: c81b6c57 Answer

Correct Answer:  
B

Rationale

Choice B is correct. Using the distributive property, the first given expression can be rewritten as  $6x^2 + 3px + 24 - 16px - 64x + 24$ , and then rewritten as  $6x^2 + (3p - 16p - 64)x + 24$ . Since the expression  $6x^2 + (3p - 16p - 64)x + 24$  is equivalent to  $6x^2 - 155x + 24$ , the coefficients of the  $x$  terms from each expression are equivalent to each other; thus  $3p - 16p - 64 = -155$ . Combining like terms gives  $-13p - 64 = -155$ . Adding 64 to both sides of the equation gives  $-13p = -91$ . Dividing both sides of the equation by  $-13$  yields  $p = 7$ .

Choice A is incorrect. If  $p = -3$ , then the first expression would be equivalent to  $6x^2 - 25x + 24$ . Choice C is incorrect. If  $p = 13$ , then the first expression would be equivalent to  $6x^2 - 233x + 24$ . Choice D is incorrect. If  $p = 155$ , then the first expression would be equivalent to  $6x^2 - 2,079x + 24$ .

Question Difficulty:  
Hard

# Question ID 5355c0ef

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 5355c0ef

$$0.36x^2 + 0.63x + 1.17$$

The given expression can be rewritten as  $a(4x^2 + 7x + 13)$ , where  $a$  is a constant. What is the value of  $a$ ?

ID: 5355c0ef Answer

Correct Answer:

.09, 9/100

Rationale

The correct answer is **.09**. It's given that the expression  $0.36x^2 + 0.63x + 1.17$  can be rewritten as  $a(4x^2 + 7x + 13)$ . Applying the distributive property to the expression  $a(4x^2 + 7x + 13)$  yields  $4ax^2 + 7ax + 13a$ . Therefore,  $0.36x^2 + 0.63x + 1.17$  can be rewritten as  $4ax^2 + 7ax + 13a$ . It follows that in the expressions  $0.36x^2 + 0.63x + 1.17$  and  $4ax^2 + 7ax + 13a$ , the coefficients of  $x^2$  are equivalent, the coefficients of  $x$  are equivalent, and the constant terms are equivalent. Therefore,  $0.36 = 4a$ ,  $0.63 = 7a$ , and  $1.17 = 13a$ . Solving any of these equations for  $a$  yields the value of  $a$ . Dividing both sides of the equation  $0.36 = 4a$  by 4 yields  $0.09 = a$ . Therefore, the value of  $a$  is **0.09**. Note that .09 and 9/100 are examples of ways to enter a correct answer.

Question Difficulty:

Hard

Question ID d8789a4c

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: d8789a4c

$$\frac{x^2 - c}{x - b}$$

In the expression above,  $b$  and  $c$  are positive integers. If the expression is equivalent to  $x + b$  and  $x \neq b$ , which of the following could be the value of  $c$  ?

- A. 4
- B. 6
- C. 8
- D. 10

ID: d8789a4c Answer

Correct Answer:

A

Rationale

Choice A is correct. If the given expression is equivalent to  $x + b$ , then  $\frac{x^2 - c}{x - b} = x + b$ , where  $x$  isn't equal to  $b$ . Multiplying both sides of this equation by  $x - b$  yields  $x^2 - c = (x + b)(x - b)$ . Since the right-hand side of this equation is in factored form for the difference of squares, the value of  $c$  must be a perfect square. Only choice A gives a perfect square for the value of  $c$ .

Choices B, C, and D are incorrect. None of these values of  $c$  produces a difference of squares. For example, when 6 is substituted for  $c$  in the given expression, the result is  $\frac{x^2 - 6}{x - b}$ . The expression  $x^2 - 6$  can't be factored with integer values, and therefore  $\frac{x^2 - 6}{x - b}$  isn't equivalent to  $x + b$ .

Question Difficulty:

Hard

# Question ID 433184f1

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 433184f1

Which expression is equivalent to  $\frac{4}{4x-5} - \frac{1}{x+1}$ ?

- A.  $\frac{1}{(x+1)(4x-5)}$
- B.  $\frac{3}{3x-6}$
- C.  $-\frac{1}{(x+1)(4x-5)}$
- D.  $\frac{9}{(x+1)(4x-5)}$

ID: 433184f1 Answer

Correct Answer:

D

### Rationale

Choice D is correct. The expression  $\frac{4}{4x-5} - \frac{1}{x+1}$  can be rewritten as  $\frac{4}{4x-5} + \frac{(-1)}{x+1}$ . To add the two terms of this expression, the terms can be rewritten with a common denominator. Since  $\frac{x+1}{x+1} = 1$ , the expression  $\frac{4}{4x-5}$  can be rewritten as  $\frac{(x+1)(4)}{(x+1)(4x-5)}$ . Since  $\frac{4x-5}{4x-5} = 1$ , the expression  $\frac{-1}{x+1}$  can be rewritten as  $\frac{(4x-5)(-1)}{(4x-5)(x+1)}$ . Therefore, the expression  $\frac{4}{4x-5} + \frac{(-1)}{x+1}$  can be rewritten as  $\frac{(x+1)(4)}{(x+1)(4x-5)} + \frac{(4x-5)(-1)}{(4x-5)(x+1)}$ , which is equivalent to  $\frac{(x+1)(4)+(4x-5)(-1)}{(x+1)(4x-5)}$ . Applying the distributive property to each term of the numerator yields  $\frac{(4x+4)+(-4x+5)}{(x+1)(4x-5)}$ , or  $\frac{(4x+(-4x))+(-4+5)}{(x+1)(4x-5)}$ . Adding like terms in the numerator yields  $\frac{9}{(x+1)(4x-5)}$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

### Question Difficulty:

Hard

Question ID ea6d05bb

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: ea6d05bb

The expression  $(3x - 23)(19x + 6)$  is equivalent to the expression  $ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. What is the value of  $b$ ?

ID: ea6d05bb Answer

Correct Answer:  
-419

Rationale

The correct answer is  $-419$ . It's given that the expression  $(3x - 23)(19x + 6)$  is equivalent to the expression  $ax^2 + bx + c$ , where  $a$ ,  $b$ , and  $c$  are constants. Applying the distributive property to the given expression,  $(3x - 23)(19x + 6)$ , yields  $(3x)(19x) + (3x)(6) - (23)(19x) - (23)(6)$ , which can be rewritten as  $57x^2 + 18x - 437x - 138$ . Combining like terms yields  $57x^2 - 419x - 138$ . Since this expression is equivalent to  $ax^2 + bx + c$ , it follows that the value of  $b$  is  $-419$ .

Question Difficulty:  
Hard

Question ID 137cc6fd

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 137cc6fd

$$\sqrt[5]{70n}\left(\sqrt[6]{70n}\right)^2$$

For what value of  $x$  is the given expression equivalent to  $(70n)^{30x}$ , where  $n > 1$ ?

ID: 137cc6fd Answer

**Correct Answer:**  
.0177, .0178, 4/225

**Rationale**

The correct answer is  $\frac{4}{225}$ . An expression of the form  $\sqrt[k]{a}$ , where  $k$  is an integer greater than 1 and  $a \geq 0$ , is equivalent to  $a^{\frac{1}{k}}$ . Therefore, the given expression, where  $n > 1$ , is equivalent to  $(70n)^{\frac{1}{5}}\left((70n)^{\frac{1}{6}}\right)^2$ . Applying properties of exponents, this expression can be rewritten as  $(70n)^{\frac{1}{5}}(70n)^{\frac{1}{6}\cdot 2}$ , or  $(70n)^{\frac{1}{5}}(70n)^{\frac{1}{3}}$ , which can be rewritten as  $(70n)^{\frac{1}{5}+\frac{1}{3}}$ , or  $(70n)^{\frac{8}{15}}$ . It's given that the expression  $\sqrt[5]{70n}\left(\sqrt[6]{70n}\right)^2$  is equivalent to  $(70n)^{30x}$ , where  $n > 1$ . It follows that  $(70n)^{\frac{8}{15}}$  is equivalent to  $(70n)^{30x}$ . Therefore,  $\frac{8}{15} = 30x$ . Dividing both sides of this equation by 30 yields  $\frac{8}{450} = x$ , or  $\frac{4}{225} = x$ . Thus, the value of  $x$  for which the given expression is equivalent to  $(70n)^{30x}$ , where  $n > 1$ , is  $\frac{4}{225}$ . Note that 4/225, .0177, .0178, 0.017, and 0.018 are examples of ways to enter a correct answer.

**Question Difficulty:**  
Hard



Question ID 34847f8a

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 34847f8a

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all  $x > 2$ , where  $r$  and  $t$  are positive constants. What is the value of  $rt$  ?

- A.  $-20$
- B.  $15$
- C.  $20$
- D.  $60$

ID: 34847f8a Answer

Correct Answer:

C

Rationale

Choice C is correct. To express the sum of the two rational expressions on the left-hand side of the equation as the single rational expression on the right-hand side of the equation, the expressions on the left-hand side must have the same denominator.

Multiplying the first expression by  $\frac{x+5}{x+5}$  results in  $\frac{2(x+5)}{(x-2)(x+5)}$ , and multiplying the second expression by  $\frac{x-2}{x-2}$  results in  $\frac{3(x-2)}{(x-2)(x+5)}$ , so the given equation can be rewritten as  $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$ , or  $\frac{2x+10}{(x-2)(x+5)} + \frac{3x-6}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$ . Since the two rational expressions on the left-hand side of the equation have the same denominator as the rational expression on the right-hand side of the equation, it follows that  $(2x+10) + (3x-6) = rx+t$ . Combining like terms on the left-hand side yields  $5x+4 = rx+t$ , so it follows that  $r = 5$  and  $t = 4$ . Therefore, the value of  $rt$  is  $(5)(4) = 20$ .

Choice A is incorrect and may result from an error when determining the sign of either  $r$  or  $t$ . Choice B is incorrect and may result from not distributing the 2 and 3 to their respective terms in  $\frac{2(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)} = \frac{rx+t}{(x-2)(x+5)}$ . Choice D is incorrect and may result from a calculation error.

Question Difficulty:

Hard

# Question ID 40c09d66

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 40c09d66

If  $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$  for all positive values of  $x$ ,  
what is the value of  $\frac{a}{b}$ ?

ID: 40c09d66 Answer

Rationale

The correct answer is  $\frac{7}{6}$ . The value of  $\frac{a}{b}$  can be found by first rewriting the left-hand side of the given equation as  $x^{\frac{\frac{5}{2}}{\frac{4}{3}}}$ . Using the properties of exponents, this expression can be rewritten as  $x^{\left(\frac{5}{2} - \frac{4}{3}\right)}$ . This expression can be rewritten by subtracting the fractions in the exponent, which yields  $x^{\frac{7}{6}}$ . Thus,  $\frac{a}{b}$  is  $\frac{7}{6}$ . Note that 7/6, 1.166, and 1.167 are examples of ways to enter a correct answer.

Question Difficulty:  
Hard

# Question ID c3b116d7

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: c3b116d7

Which of the following expressions is(are) a factor of  $3x^2 + 20x - 63$ ?

- I.  $x - 9$

II.  $3x - 7$
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: c3b116d7 Answer

Correct Answer:  
B

Rationale

Choice B is correct. The given expression can be factored by first finding two values whose sum is **20** and whose product is **3(−63)**, or **−189**. Those two values are **27** and **−7**. It follows that the given expression can be rewritten as  **$3x^2 + 27x - 7x - 63$** . Since the first two terms of this expression have a common factor of  **$3x$**  and the last two terms of this expression have a common factor of **−7**, this expression can be rewritten as  **$3x(x + 9) - 7(x + 9)$** . Since the two terms of this expression have a common factor of  **$(x + 9)$** , it can be rewritten as  **$(3x - 7)(x + 9)$** . Therefore, expression II,  **$3x - 7$** , is a factor of  **$3x^2 + 20x - 63$** , but expression I,  **$x - 9$** , is not a factor of  **$3x^2 + 20x - 63$** .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:  
Hard

Question ID 371cbf6b

| Assessment | Test | Domain        | Skill                  | Difficulty                                   |
|------------|------|---------------|------------------------|----------------------------------------------|
| SAT        | Math | Advanced Math | Equivalent expressions | <div><div></div><div></div><div></div></div> |

ID: 371cbf6b

$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$

The equation above is true for all  $x$ , where  $a$  and  $b$  are constants. What is the value of  $ab$  ?

- A. 18
- B. 20
- C. 24
- D. 40

ID: 371cbf6b Answer

Correct Answer:  
C

Rationale

Choice C is correct. If the equation is true for all  $x$ , then the expressions on both sides of the equation will be equivalent. Multiplying the polynomials on the left-hand side of the equation gives  $5ax^3 - abx^2 + 4ax + 15x^2 - 3bx + 12$ . On the right-hand side of the equation, the only  $x^2$ -term is  $-9x^2$ . Since the expressions on both sides of the equation are equivalent, it follows that  $-abx^2 + 15x^2 = -9x^2$ , which can be rewritten as  $(-ab + 15)x^2 = -9x^2$ . Therefore,  $-ab + 15 = -9$ , which gives  $ab = 24$ .

Choice A is incorrect. If  $ab = 18$ , then the coefficient of  $x^2$  on the left-hand side of the equation would be  $-18 + 15 = -3$ , which doesn't equal the coefficient of  $x^2$ ,  $-9$ , on the right-hand side. Choice B is incorrect. If  $ab = 20$ , then the coefficient of  $x^2$  on the left-hand side of the equation would be  $-20 + 15 = -5$ , which doesn't equal the coefficient of  $x^2$ ,  $-9$ , on the right-hand side.

Choice D is incorrect. If  $ab = 40$ , then the coefficient of  $x^2$  on the left-hand side of the equation would be  $-40 + 15 = -25$ , which doesn't equal the coefficient of  $x^2$ ,  $-9$ , on the right-hand side.

Question Difficulty:  
Hard